

Grade 8
Unit 12
Lesson 9 Practice

Name Answer Key
Date _____
Period _____

1. The following formula is used to calculate a patient's Body Mass Index (BMI), where w is a patient's weight in pounds (lb) and h is a patient's height in inches (in.).

$$\frac{w}{h^2} \cdot 703$$

Determine the BMI for an adult who is 6 feet 2 inches tall and weighs 205 pounds. Round your answer to the nearest tenth. $6'2" = 6(12) + 2 = 74 \text{ in.}$

$$w = 205 \qquad \frac{205}{74^2} \cdot 703 \rightarrow \frac{205}{5,476} \cdot 703 \approx \boxed{26.3}$$

$$h = 74$$

2. The surface area (SA) of a cone can be found using the formula $SA = \pi \cdot r \cdot \sqrt{r^2 + h^2}$, where r is the radius of the cone and h is the height of the cone. The expression shown can be used to find the surface area of a sugar cone with a radius of 2.75 centimeters (cm) and a height of 12 cm.

$$\pi \cdot 2.75 \cdot \sqrt{2.75^2 + 12^2}$$

Find the surface area of the sugar cone in square cm. Use 3.14 to approximate π .

$$3.14 \cdot 2.75 \cdot \sqrt{7.5625 + 144} \rightarrow 3.14 \cdot 2.75 \cdot \sqrt{151.5625} \\ \approx \boxed{106.3 \text{ cm}^2}$$

3. Television screen dimensions are advertised using the measure of the diagonal distance across the screen. Paul plans to buy a 55-inch television that measures 47.9 inches in width. Paul uses the expression shown to find the height of the space he needs to hang the television so there is at least 4 inches above and below the screen.

$$\sqrt{55^2 - 47.9^2} + 2(4)$$

Find the height of the space Paul needs. Round to the nearest hundredth.

$$\sqrt{3025 - 2294.41} + 8 \approx \boxed{35.05 \text{ in.}}$$

4. Kinetic energy can be calculated using the expression $\frac{1}{2}mv^2$, where m is the mass of an object and v is the object's velocity. If a cyclist has a mass of 73 kilograms (kg), his bicycle has a mass of 24 kg, and he is traveling at a speed of 9 meters (m) per second, the expression $\frac{1}{2}(73 + 24)(9)^2$ can be used to find the kinetic energy stored, measured in Joules. Calculate the kinetic energy.

$$\frac{1}{2}(97)(81) = \boxed{3,928.5 \text{ Joules}}$$

5. Dorian needs to move some of his possessions into a storage unit. He has 5 cubic boxes that measure 24 inches (in.) on each side. He also has 4 cubic boxes that measure 20 in. on each side. To determine the amount of space all the boxes will take up, Dorian uses the equation $5 \cdot 24^3 + 4 \cdot 20^3$. How many cubic inches of storage does Dorian need?

$$5 \cdot 13,824 + 4 \cdot 8,000 = 69,120 + 32,000 = \boxed{101,120 \text{ in.}^3}$$

For questions 6-10, find the value of each expression.

6. $\left(\frac{1}{2}\right)^3 \cdot (\sqrt{36} + 5)$ $\frac{1}{8} \cdot (6 + 5) \rightarrow \frac{1}{8} \cdot 11 = \boxed{\frac{11}{8} \text{ or } 1\frac{3}{8}}$

7. $(7^2 + \sqrt{121}) - 6.75$ $(49 + 11) - 6.75 \rightarrow 60 - 6.75 = \boxed{53.25}$

8. $\left(\frac{1}{3}\right)^{-2} \cdot (\sqrt{16} + 6)^2$ $\left(\frac{3}{1}\right)^2 \cdot (\sqrt{16} + 6)^2 \rightarrow 9 \cdot 10^2 \rightarrow 9 \cdot 100 = \boxed{900}$

9. $\left(\frac{1}{4}\right)^2 \cdot (\sqrt[3]{-64} + 6^2)$ $\frac{1}{16} \cdot (-4 + 36) \rightarrow \frac{1}{16} \cdot \frac{32}{1} = \boxed{2}$

10. $(2)^{-2} \cdot \left(\sqrt[3]{\frac{1}{125}} + 6\right)$

$$\left(\frac{1}{2}\right)^2 \cdot \left(\sqrt[3]{\frac{1}{125}} + 6\right) \rightarrow \frac{1}{4} \cdot \left(\frac{1}{5} + 6\right) \rightarrow \frac{1}{4} \cdot \left(\frac{1}{5} + \frac{30}{5}\right) \rightarrow \frac{1}{4} \cdot \frac{31}{5} = \boxed{\frac{31}{20} \text{ or } 1\frac{11}{20}}$$